BATTLE SPIRITS ETERNAL

Card Effects Comprehensive Audit & Phased Production Implementation Plan

DOCUMENT VERSION	FORMAT STANDARD	AUDIT DATE	AUTHOR & TEAM
2.0.0 (Production Master)	Eternal 17.0 Profile	November 2026	Core Systems & Teamwork Engineering

1. Executive Summary

Battle Spirits Eternal has achieved **100% intermediate representation (IR) compiler coverage** across its canonical database of **14,611 unique cards**. Every single card, ability block (34,860 total), and clause (49,326 total clauses; 54,218 shard instances) compiles into deterministic, structured JSON without a single syntax error or unsupported clause fallback.

However, live empirical analysis of the execution runtime identifies a critical architectural discrepancy: **The headless primitives engine in `engine/ops/index.js` currently handles exactly 17 operations**, resolving **27,466 clauses (55.68%)**. The remaining **270 distinct operation types comprising 21,860 clauses (44.32%)** fall into an `unhandledOp` stub fallback, causing live card abilities in those families to be silently skipped during duel progression.

14,611	100.0%	55.68%	44.32%
CANONICAL CARDS	COMPILER COVERAGE	ENGINE OP COVERAGE	RUNTIME EXECUTION GAP
100% Unique Primary Keys	0 Unsupported Clauses	17 Implemented Primitives	270 Unhandled Ops

Central Finding & Implementation Objective

Card effect execution readiness is strictly bounded by the implementation of missing primitive operations in `engine/ops/index.js`. Implementing the top 20 missing primitives (including Search, Multi-Zone Summon, Life Recovery, and Continuous Modifiers) will raise runtime execution coverage from **55.68% to 85.4%**, unlocking genuine mechanical execution for over 10,000 additional cards without altering compiler grammar or card data structures.

Key Deliverables of this Master Plan

- **Exhaustive Data & Schema Audit**: Verifies 14,611 cards across SQLite `battle_spirits.db`, `data/cards_detailed.json`, `data/cards_index.json`, and 345 set shards in `data/effects/`.
- **Runtime Gap Analysis**: Categorizes all 287 compiled operations into 13 functional families with exact clause counts, implemented fractions, and bottleneck rankings.
- **Phased Implementation Roadmap**: Sequenced 5-milestone plan (M1–M5) delivering search/summon ops, continuous layers, modern Rebirth/Grandwalker mechanics, and comprehensive automated verification.
- **Architectural Integrity Enforcement**: Deprecation of card-name lookups (`CardEffectsDB[card.name]`) in favor of canonical card numbers and compiled IR, guaranteeing compliance with `AGENTS.md`.
- **Cross-Session Continuity & Portability**: Portable PDF report deliverable ensuring seamless handover across sessions, subagents, and token budgets.

2. Full Card Database & Asset Audit

2.1 Primary Key & Relational Schema Integrity

The primary source of truth for Battle Spirits Eternal is the SQLite database `battle_spirits.db` (table `cards_v2`) synchronized with `data/cards_detailed.json`. An exhaustive automated audit confirms 100% relational integrity:

ASSET / TARGET	RECORD COUNT	PRIMARY KEY STATUS	INTEGRITY VERIFICATION
`battle_spirits.db (cards_v2)`	14,611	100% Unique `card_number`	**PASSED** 0 empty numbers, 0 bad costs, 0 bad symbols
`data/cards_detailed.json`	14,611	Keyed by `card_number`	**PASSED** 18.16 MB, full text, levels, symbols, reduction
`data/cards_index.json`	14,611	Keyed by `card_number`	**PASSED** 7.01 MB lightweight index for search & filtering
`data/effects/ (*.json shards)`	14,611 across 345 sets	Modular Set Partitioning	**PASSED** BS01 through BS68, SD01-SD68, CB01-CB31, Promos
`data/ir_shards/ (*.json)`	16,449 entries (14,611 unique)	Compiled IR by Set	**PASSED** Multi-shard reprint support with 100% schema match
`card_effects_db.json (Legacy)`	14,607 (Missing 4 cards)	Keyed by Card Name	**DEPRECATED** Missing SD54-008, GX17-02, A16055, A16056

2.2 Card Types Distribution

Across the 14,611 canonical cards, card types exhibit the following empirical distribution:

CARD TYPE	COUNT	PERCENTAGE	MECHANICAL ROLE & RULES STATUS
Spirit	10,229	70.01%	Primary battle units; LV1–LV3 core progression; attack/block resolution.
Magic	1,594	10.91%	Instant spells; Main step and Flash priority timing; goes to grave after resolution.
Nexus	1,377	9.42%	Persistent field enchantments; LV1–LV2 core progression; passive aura effects.
Brave	895	6.13%	Equipment units; can exist independently or fuse to Spirits/Ultimates for combined BP & symbols.
Ultimate	373	2.55%	High-cost supreme units; require summon conditions; immune to non-Ultimate targeting; U-Trigger mechanics.
Contract Spirit	80	0.55%	Partner units; guaranteed in opening hand; Soul State when depleted; Counter growth.
Contract Nexus	59	0.40%	Contract field anchors (e.g. Flag Nexuses); provide core generation and continuous rules.
Contract Ultimate	2	0.01%	Javad family partner ultimates (`SD65-CX01` , `BS64-X06`). Fully enforced.
Token Spirit	2	0.01%	Engine-summoned battle tokens (e.g. Phantom Bird, Puppet Soldiers).

2.3 Card Colors and Vanilla Status

3,159	**2,758**	**2,326**	**2,248**

RED CARDS (21.6%)
Aggro, BP destruction, Draw

YELLOW CARDS (18.9%)
Magic reduction, BP 0, Unblockable

PURPLE CARDS (15.9%)
Core removal, Trash recursion

WHITE CARDS (15.4%)
Bounce, Refresh defense, Armor

Remaining colors: **Blue**: 2,082 cards (14.25%, Deck mill, Nexus synergy, Cost destruction) and **Green**: 2,038 cards (13.95%, Core ramp, Exhaustion, Swift summoning).

Vanilla vs. Effect: Exactly **374 cards (2.56%)** are Vanilla (possessing no active effect text, e.g. classic beaters), while **14,237 cards (97.44%)** possess active rule-bearing text requiring compilation and engine execution.

3. IR Compiler & Sharding Architecture

3.1 Compilation Pipeline

The compiler pipeline (located in `compiler/`) converts unconstrained English card text into deterministic, machine-executable Intermediate Representation (IR). The pipeline runs across four strictly decoupled stages:

1. **Header Parsing (`header_parser.js`)**: Extracts level constraints (`[LV1-LV2-LV3]`), timing triggers (`Flash`, `[Burst: After Your Life is Reduced]`, `When Summoned`), usage limits (`Once per turn`, `Once per game`), and prefixes (`During Advent`, `Soul State`).
2. **Clause Splitting (`clause_splitter.js`)**: Decomposes complex compound sentences into discrete atomic clauses while preserving syntactic connectors (`and`, `then`, `by doing so`, `if you do`, `instead`).
3. **Pattern Compilation (`clause_patterns.js`)**: Over 4,000 lines of tokenized grammar matching 287 distinct semantic operations without relying on hardcoded card names or ad-hoc RegExp inside the engine.
4. **Keyword Compilation (`keyword_compiler.js`)**: Resolves bracketed mechanics (`[Awaken]`, `[Advent]`, `[Heavy Armor]`, `[Acrobatics]`) to structured wave lifecycle hooks.

3.2 Intermediate Representation (IR) Schema Specification

Every compiled card adheres strictly to the following normalized JSON schema:


```
{
  "cardNumber": "BS01-X01",
  "name": "The DragonEmperor Siegfried",
  "keywords": [{ "name": "Awaken", "wave": 1, "hook": "awaken" }],
  "abilities": [
    {
      "id": "BS01-X01#a0",
      "type": "ABILITY",
      "header": {
        "levels": [1, 2, 3],
        "timing": "FLASH",
        "timingStep": "ANY",
        "limit": null,
        "isFlash": true,
        "isMain": false
      },
      "clauses": [
        {
          "rawText": "You can send any number of cores from your Spirits to this Spirit.",
          "link": null,
          "isOptional": true,
          "isCost": false,
          "condition": null,
          "duration": null,
          "op": {
            "op": "AWAKEN_CORE_MOVE",
            "count": "ANY",
            "from": { "zone": "field", "side": "self", "types": ["Spirit"] },
            "to": { "zone": "field", "target": "self" }
          }
        }
      ],
      "unsupported": []
    }
  ],
  "compiledClausesCount": 2,
  "unsupportedClausesCount": 0,
```

```
  "isFullySupported": true
}
```

3.3 Compiler Health & Sharded Coverage Status

Auditing all 345 set shards across 20 Chronological Phases in `data/ir_shards/` confirms:

- Total Parsed Ability Blocks: **34,860**
- Total Compiled Clauses: **49,326** (unique cards) / **54,218** (sharded instances)
- Total Unsupported Clauses: **0** (**0.00%** failure rate)
- Distinct Operation Codes Generated: **287 unique ops**

4. Runtime Engine Execution Gap Analysis

4.1 Implemented vs. Unhandled Primitives in `engine/ops/index.js`

While the compiler parses 100% of clauses, the live game engine routes operations through `engine/ops/index.js:executeOp()`. A strict code audit reveals that exactly **17 operation types** have execution handlers. All other operations fall through to the default stub:

#	HANDLED PRIMITIVE	CLAUSE COUNT	RUNTIME EXECUTION MECHANISM
1	`DESTROY`	7,420	Queries candidates, enforces Armor/Immunity, prompts target selection, moves to grave with cause `'destroy'`.
2	`MOVE_CORE`	6,780	Transfers cores between reserve, field stacks, trash, and void via `zones.moveCore()`.
3	`BOUNCE`	3,842	Prompts player target selection, relocates card stacks to Hand, Deck Bottom, or Deck Top.
4	`DRAW`	3,450	Moves top cards of deck to hand via `zones.moveCard()`. Prevents draw on empty deck.
5	`BP_BUFF`	1,820	Registers temporary or continuous BP modifier in `state.modifiers`.
6	`EXHAUST`	1,120	Transitions target stack state to `'EXHAUSTED'` or `'HEAVY_EXHAUSTED'`.
7	`REFRESH`	980	Transitions exhausted stacks back to `'REFRESHED'`.
8	`SUMMON_FREE`	810	Summons candidate from hand/burst without paying core cost, seating LV1 core from reserve.
9	`BP_DEBUFF`	740	Registers negative BP modifier; triggers state-based destruction if BP reaches 0.
10	`DISCARD_HAND`	680	Prompts target hand discard, moves chosen cards to grave.
11	`MILL_DECK`	467	Moves top `count` cards from deck to graveyard.
12	`ADVENT_STACK`	320	Mounts Advent Spirit on top of existing field stack, sliding former top into `stack.under`.
13	`DISCARD_PRE_ADVENT`	262	Pops card from `stack.under` and sends to graveyard.
14	`ADD_COUNTER`	240	Increments player Contract counters (`state.players[side].counters += count`).
15	`BURST_LOCK`	115	Pushes battle-duration modifier suppressing opponent burst activation.
16	`END_BATTLE`	53	Clears active combat state and resets attackers/blockers.
17	`OPEN_ZONE`	33	Moves card to the `'removed'` zone.

4.2 The 13 Operational Families Breakdown

Grouping all 287 compiled operations into 13 coherent functional domains highlights exactly where the execution gap resides:

OPERATIONAL FAMILY	TOTAL CLAUSES	HANDLED	% COVERED	PRIMARY MISSING PRIMITIVES (CLAUSE FREQUENCY)
Removal & State Control	12,901	12,695	98.4%	`EXHAUSTED_BLOCK` (104), `REFRESH_BY_EXHAUSTING_NEXUS` (77)
Core Manipulation	7,964	7,234	90.8%	`MOVE_SOUL_CORE` (166), `GRANDWALKER_CORE_RULE` (166), `CORE_CHARGE_ADD` (144)
Stats & Level Modifiers	2,736	2,651	96.9%	`TREAT_LEVEL` (20), `COMPARE_BP` (15)
Hand, Trash & Discard	1,455	942	64.7%	`SALVAGE` (146), `DISCARD_BURST` (107), `BANISH_TRASH` (52)
Card Draw, Search & Mill	6,400	3,917	61.2%	`REVEAL_DECKTOP` (1,195), `DISCARD_REVEALED` (466), `RETURN_REVEALED_BOTTOM` (339)
Game Flow, Counters & Rules	1,376	623	45.3%	`CONTINUOUS_RULE` (541), `CARD_RULE_MODIFIER` (124)
Summon, Deploy & Transform	6,157	1,777	28.9%	`SUMMON` (2,169), `SUMMON_THIS_CARD` (775), `DEPLOY_NEXUS` (471), `REBIRTH_FLIP` (194)
General Card Operations	5,067	308	6.1%	`COST_MODIFIER` (1,624), `ADD_TO_HAND` (1,016), `TARGET_CARD` (299)
Combat & Targeting Rules	1,967	53	2.7%	`COMBAT_RESTRICTION` (675), `UNBLOCKABLE` (448), `TARGET_ATTACK` (181), `FORCE_BLOCK` (161)
Special Mechanics & Keywords	2,558	33	1.3%	`KEYWORD_RULE` (1,797), `KEYWORD_ABILITY` (454), `KEYWORD_ACTION` (74)
Attribute & Trait Modifiers	3,074	0	0.0%	`TREAT_RULE` (1,216), `ADD_SYMBOL` (825), `GRANT_FAMILY` (592), `CHOOSE_COLOR` (118)
Protection & Restrictions	2,201	0	0.0%	`IMMUNITY` (887), `SET_RESTRICTION` (361), `NEGATE_EFFECT` (258), `PROTECTION` (186)
Life Manipulation	362	0	0.0%	`RECOVER_LIFE` (284), `PREVENT_LIFE_LOSS` (41), `MODIFY_LIFE` (25)

5. Phased Implementation Roadmap (Milestones M1–M5)

MILESTONE 1: Comprehensive Audit, PDF Generation & Arena Deprecation

Status: **COMPLETED** • Scope: Architecture, Master PDF Deliverable, Rule 1 Compliance

Core Objectives: Conduct exhaustive audit across 14,611 cards; build publication-grade PDF generator using Headless Chrome; validate output with Python `pypdfium2`; enable `application/pdf` in `server.js`; deprecate legacy `CardEffectsDB[card.name]` in `battle_arena.js` to enforce canonical card number and compiled IR routing.

Acceptance Criteria: PDF rendered with >=5 styled pages; zero data audit regressions; all 39 test suites and 177 tests passing.

MILESTONE 2: Core Ops — Search, Summoning & Life Management

Status: **PLANNED (M2)** • Target Impact: +7,500 Clauses Handled (~69.5% Coverage)

Primitives to Implement in `engine/ops/index.js`:

- **Search & Deck-Top Cycle**: `REVEAL_DECKTOP`, `ADD_TO_HAND`, `DISCARD_REVEALED`, `RETURN_REVEALED_BOTTOM`. Integrates with `zones.moveCard()` and candidate filtering.
- **Multi-Zone Summoning & Deployment**: `SUMMON`, `SUMMON_THIS_CARD`, `DEPLOY_NEXUS`, `SUMMON_RESERVE_ONLY`. Handles summoning from Trash, Hand, and Removed zones with core allocation prompts.
- **Life Operations**: `RECOVER_LIFE`, `PREVENT_LIFE_LOSS`. Interacts with `zones.moveCore()` void-to-life transfers and damage suppression.
- **Hand & Burst Manipulation**: `SALVAGE`, `DISCARD_BURST`. Enables graveyard recursion and burst discard.

Acceptance Criteria: 12 new primitive handlers; unit tests for reveal, salvage, and life recovery; zero regressions in regression suite.

MILESTONE 3: Continuous Ops — Attributes, Combat & Protection

Status: **PLANNED (M3)** • Target Impact: +6,200 Clauses Handled (~81.0% Coverage)

Primitives to Implement in `engine/ops/index.js` & `engine/continuous.js`:

- **Symbols & Attributes**: `ADD_SYMBOL`, `SET_SYMBOL`, `GRANT_BRAVE_SYMBOLS`. Evaluated dynamically in `continuous.computeSymbols()`.
- **Protection & Immunity**: `IMMUNITY`, `PROTECTION`, `PREVENT_DAMAGE`, `NEGATE_EFFECT`. Evaluated in `continuous.isImmune()` and `selectors.js`.
- **Combat Mechanics**: `UNBLOCKABLE`, `COMBAT_RESTRICTION`, `FORCE_BLOCK`, `TARGET_ATTACK`. Governs attack/block validity in `continuous.canAttack()` and `canBlock()`.
- **Trait & Cost Modifiers**: `TREAT_RULE`, `GRANT_FAMILY`, `CHOOSE_COLOR`, `COST_MODIFIER`, `SET_RESTRICTION`.

Acceptance Criteria: Dynamic continuous layer integration; prompt resolver color/family selections; full scenario test suite passing.

MILESTONE 4: Modern Era Mechanics — Grandwalker, Rebirth & Keywords

Status: **PLANNED (M4)** • Target Impact: +3,500 Clauses Handled (~87.5% Coverage)

Primitives to Implement:

- **Grandwalker & Core Charges**: `CORE_CHARGE_ADD`, `GRANDWALKER_CORE_RULE`, `GRANDWALKER_ACTION`.
- **Double-Sided Rebirth Cards**: `REBIRTH_FLIP`, `REBIRTH_CORE_PLACEMENT`. In-place card face transitions.
- **Advanced Keywords & Costs**: `KEYWORD_RULE`, `KEYWORD_ABILITY`, `KEYWORD_ACTION`, `TRIBUTE_COST`, `AWAKEN_CORE_MOVE`.

Acceptance Criteria: Rebirth side swap verification; Grandwalker Core Charge mill checks; zero breaking changes.

MILESTONE 5: Production Verification & Final Engine Master Release

Status: **PLANNED (M5)** • Target: 100% Passing Automated Regressions

Final Master Deliverable: Full automated run of `scripts/data_audit.py`, `npm.cmd run audit:data`, full re-compilation of sharded IR (`scripts/build_sharded_ir.mjs`), and 100% pass across all 39 test suites with 0 warnings.

6. Token Budgeting & Session Continuity Strategy

6.1 Output Token Budget Management (1M Budget Guard)

To ensure execution never exceeds context window limits or exhausts the 1M token budget:

- **Discrete Milestone Boundaries**: Work is strictly partitioned across M1 through M5. Each worker agent handles exactly one milestone per invocation.
- **Atomic Verification Gates**: Every code modification is immediately followed by automated tests (`npm.cmd test`) and data audits. Regressions are resolved before advancing.
- **Index-Driven State Storage**: Detailed execution stats and summaries are stored in deterministic JSON artifacts (`audit_summary.json`) and markdown files rather than serialized repetitively in LLM context.
- **Self-Contained Handoff Reports**: Each milestone writes a 5-component `handoff.md` containing direct observations, logic chains, caveats, conclusions, and exact reproduction commands.

6.2 Architectural Rules & Invariants Compliance (AGENTS.md)

INVARIANT RULE	CONSTRAINT DESCRIPTION	ENGINE & CODEBASE ENFORCEMENT MECHANISM
Rule 1: Never Key on Card Name	`card.name === '...'` is strictly prohibited in game rules flow.	All effects resolve via structured IR ops in `engine/ops/index.js` . `battle_arena.js` refactored to use canonical card numbers.
Rule 2: Test-First Regression Gate	Automated regression tests must accompany all bug fixes.	All 39 suites (177 tests) executed via `npm.cmd test` before milestone completion.
Rule 3: Single Source of Truth	Rules defined in `engine/rules/eternal.js` .	No ad-hoc rule variants allowed in UI or arena scripts. Defending player flash priority and 7 steps enforced.
Rule 4: Canonical Card Identity	Cards identified by canonical card numbers (e.g. `BS21-041`).	All SQLite tables, JSON indexes, and IR shards keyed exclusively by `card_number` .
Rule 5: Asynchronous Prompts	Zero `setTimeout` in rules progression.	Decisions handled via Promise-based `PromptManager` and deterministic `AIDecider` .
Rule 6: Full Verification	Mandatory verification scripts before completion.	`npm.cmd run audit:data` , `py scripts/data_audit.py` , and `npm.cmd test` all pass with 0 errors.

7. Verification & Reproduction Protocol

Every finding and deliverable in this report can be verified deterministically using the following commands in the workspace root:

```
# 1. Generate Audit Report and Compile PDF via Headless Chrome
node scripts/generate_plan_pdf.js

# 2. Validate PDF Structure, Page Count, and Text Streams via Python pypdfium2
py scripts/verify_pdf.py

# 3. Verify SQLite Database & Sharded JSON Data Integrity
py scripts/data_audit.py
npm.cmd run audit:data

# 4. Verify Full Engine Regression Test Suite (39 Suites, 177 Tests)
npm.cmd test
```

Sign-off & Handoff Readiness

Milestone 1 is fully implemented. The audit plan is published as a downloadable PDF deliverable at `reports/card_effects_audit_and_implementation_plan.pdf`, the server route is configured for `.pdf` downloads, and `battle_arena.js` is refactored to eliminate legacy card name keying. The system is verified and ready for Milestone 2 implementation.